

Afia Sultana

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<https://github.com/afia2002>

Career Objective

Motivated Computer Science graduate seeking an internship opportunity to apply my skills in Software Development, Machine Learning, Data Analysis, Automation, Modern Testing, and Web Development. Strong in problem-solving, analytical thinking, and teamwork, with a keen interest in research and data-driven solutions. Passionate about leveraging Artificial Intelligence, Machine Learning, and computer science principles to solve real-world challenges while continuously learning and growing professionally.

Education

Bachelor of Computer Science and Engineering American International University-Bangladesh, Dhaka	Expected Graduation Year: 2026
Higher Secondary Certificate (HSC) Dhaka Cantonment Girls' Public School and College, Dhaka Science	2018-2020
Secondary School Certificate (SSC) Monipur Uchcha Vidyalaya and College, Dhaka Science	2017-2018

Academic Projects

1. In Search of Titanic

[\(Github link\)](#)

Tools/Technologies Used: Computer Graphics, C/C++ , OpenGL, GLUT

Description: Created a visual simulation project that represents an underwater journey to locate the Titanic using basic computer graphics techniques, focusing on object animation, scene rendering, and user interaction to enhance visual storytelling.

2. Anti-Money Laundering Using Graph Attention Networks (GAT)

[\(GitHub Link\)](#)

Tools/Technologies Used: Python, Machine Learning, Graph Attention Networks

Description: Implemented a machine learning model using Graph Attention Networks to detect suspicious financial transactions, aiming to improve accuracy in identifying money laundering activities by analyzing transaction patterns and relationships.

3. Virtual Reality Therapy For Phobias

[\(GitHub Link\)](#)

Tools/Technologies Used: Figma, Microsoft Word

Description: Developed a software solution using virtual reality to help users gradually overcome phobias through immersive exposure therapy, combining psychology principles with interactive 3D environments to create a safe and controlled treatment experience.

4. Weather Classification using YOLOv8

[\(GitHub Link\)](#)

Tools/Technologies Used: Python, YOLOv8, Computer Vision, Deep Learning

Description: Built a computer-vision pipeline using Ultralytics YOLOv8 and PyTorch, preparing a 5-class image dataset . Implemented a CBAM attention block and integrated it into the YOLOv8 backbone to compare baseline vs attention-enhanced performance.

5. Employee Management System

[\(GitHub Link\)](#)

Tools/Technologies Used: Python, SQLite, HTML, CSS, Django, Web Development, Database Management

Description: Developed a Django-based Employee Management System to manage employee records with features for adding, viewing, updating, and deleting employee information.

Research works

1.Predicting Future Air Quality and Assessing Its Impacts in Dhaka: Urbanization, Sources, and Mitigation Strategies

[\(Github Link\)](#)

▪ This research predicts future air quality in Dhaka by using machine learning models such as Random Forest, XGBoost, and LSTM. It uses air quality, weather, and urbanization data collected from 2016 to 2022 to predict daily AQI. Random Forest gave the best results among the models. The study also identifies pollution sources and suggests ways to reduce air pollution in Dhaka.

2.Exploring-Data-Mining-Algorithms-for-Effective-Financial-Fraud-Detection

[\(GitHub Link\)](#)

▪ Proposed a hybrid financial fraud detection framework integrating Genetic Algorithm-based feature selection, semi-supervised SVM, Random Forest + XGBoost ensemble, and Isolation Forest, achieving 0.87 ROC-AUC while effectively handling highly imbalanced transaction data

Skills & Abilities

- **Programming Languages:** Python, HTML, C++, R , CSS, PHP, Java-Script
- **Tools & Software:** Microsoft Excel, Word , PowerPoint , Canva , AUTO-CAD, Cisco Packet Trace, emu8086.
- **Language :** English, Bangla

Certificate

Automation and Modern Testing Tools : [\[View Certificate\]](#)